

Kamal Joshi

Senior System Software Engineer — Data Platform & Observability | Go | Kubernetes | Terraform | Distributed Systems

Austin, TX | Open to remote | Open to relocation for the right opportunity

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PROFESSIONAL SUMMARY

Senior system software engineer and technical leader with deep experience building high-performance distributed platforms, telemetry pipelines, data ingestion systems, and cloud-native infrastructure. Proven track record of full-stack platform ownership — from backend ingestion architecture and API design through policy enforcement, credential management, audit logging, and production incident response. Strong hands-on experience with Go, Kubernetes, Terraform, event-driven pipelines, and time-series databases (OpenTSDB). Led ISMAP compliance certification and built governance controls into production platforms. Over the last two years I have gotten genuinely effective at directing AI tools (GitHub Copilot, Claude Code, Codex) to produce safe, testable, readable code across the full SDLC while owning the design, quality bar, and business outcome.

CORE STRENGTHS

- **Observability & telemetry pipelines:** time-series databases (OpenTSDB), metrics instrumentation, event ingestion at scale, DLQ recovery, push-based delivery architectures, Kinesis Firehose, S3/Iceberg data lake formats
- **Infrastructure-as-Code:** Terraform modules and pipelines, multi-account/multi-region infrastructure, Kubernetes (EKS), Docker, CI/CD
- **Policy enforcement & data governance:** OPA (Open Policy Agent), OAuth 2.0, OIDC, AWS Secrets Manager, audit logging, secure credential management; led ISMAP security certification
- **Backend platform engineering:** Go, Java, Spring Boot, gRPC, REST/JSON APIs, webhooks, CloudEvents, event-driven microservices, SQL
- **Streaming & storage:** Kinesis Firehose, SQS/SNS, DynamoDB streams, RabbitMQ, S3/Iceberg; analyzed multi-region failover for pipelines spanning EMR on EKS and downstream delivery
- **Full-stack fluency:** TypeScript, JavaScript, Angular; designed and shipped developer-facing dashboards and API-driven UIs
- **AI-assisted SDLC:** daily use of Claude Code, GitHub Copilot, and Codex across implementation, IaC, testing, debugging, and design exploration

EXPERIENCE

Cisco

Software Engineering Technical Leader

November 2025 - Present | Austin, TX

- Architected and owned centralized telemetry and event-delivery pipelines for Cisco Secure Access, handling high-volume security events across multi-region, sovereign, and GovCloud environments — including Kinesis Firehose, S3/Iceberg, EMR on EKS, SQS/SNS, DynamoDB, and webhook delivery.
- Prevented security-event data loss during a production incident by identifying a **300k+ message DLQ backlog**, quantifying impact across **15 organizations**, and executing a full redrive within **5-10 minutes** — zero message loss.
- Drove integrations-svc production readiness across **sovereign and multi-region environments** by decomposing a high-risk Terraform rollout into staged changes across ECR, Secrets Manager, DynamoDB streams, SNS, service config, and audit logging, while validating HA behavior across at least two regions.
- Built policy enforcement and secure credential infrastructure: mapped AuthSvc TLS secret synchronization, API key refresh behavior, and access control flows across Kubernetes, Secrets Manager, and Kong — reducing ambiguity for security-sensitive automation.

- Led cross-service design for customer-selected **Splunk** and **CloudEvents** delivery formats across alerts, security events, integrations, and notifications services — unifying distinct observability signals into a consistent pipeline.
- Led design and delivery of push security events capability using the open **CloudEvents** standard, shipping **alpha on schedule (Nov 2025)** with a team of three engineers.
- Resolved a customer-scale onboarding failure by tracing **250+ appliance registration requests** through API key auth and rate-limit behavior, driving durable fixes around retry/backoff, token reuse, and service ownership.

Senior Software Engineer

January 2021 - November 2025 | Austin, TX

- Implemented **metrics instrumentation** for Authorization Service and Integrations Service, adding structured telemetry across key platform components.
- Deployed **OPA (Open Policy Agent)** as a sidecar for authorization policy enforcement, building governance controls directly into platform services rather than administering them externally.
- Led **ISMAP security certification** for the APIX platform, coordinating cross-functional compliance requirements and audit evidence — end-to-end ownership from scoping through sign-off.
- Built auth-tools, an internal authorization library adopted across multiple platform services to simplify OAuth/policy enforcement integration.
- Pioneered team-wide **GitHub Copilot** adoption ahead of broader Cisco AI initiatives; integrated AI tooling (Claude Code, Codex) into daily workflows across IaC authoring, test generation, debugging, and code review.

Randstad Technologies US at Cisco

Senior Software Developer

March 2015 - January 2021 | Dublin, OH

- Designed and implemented **Go-based gRPC and JSON/HTTP APIs** for a cloud-native **data ingestion platform**; used **OpenTSDB** as the time-series database for platform metrics (~2017–2019).
- Built API-driven authentication and authorization flows using OIDC, OAuth 2.0, and gRPC-Web; contributed to automatic generation of protocol buffer specs from SQL schemas.
- Designed a type-assisted search feature for an Active Threat Analytics dashboard, reducing friction for security engineers writing threat-hunting queries against large data sets.
- Reduced **100+ OWASP-reported security warnings to zero** through secure coding, browser security headers, input sanitization, and server-side validation.

Earlier Engineering Experience

Cognizant, Oracle, Collabera | December 2007 - February 2015

- Delivered backend services, workflow automation, and integration features across financial services, enterprise content, healthcare, insurance, and public-sector domains.
- Built a foundation in Java, REST APIs, service integrations, data pipelines, and maintainable engineering practices.

CERTIFICATIONS

- AWS Certified Solutions Architect - Associate
- AWS Certified Developer - Associate
- MongoDB Certified Java Developer M101J
- Google Cloud Platform Essentials

EDUCATION

Bachelor of Engineering, Information Technology
Dharmsinh Desai University